KAUNG KHANT ZAW

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EDUCATION

Bachelor of Engineering in Computer Engineering — **Bangkok, Thailand**
King Mongkut's Institute of Technology Ladkrabang (KMITL) — **2022-2026**
- GPA: 3.62/4.0

Computer Science Coursework — **Yangon, Myanmar**
University of Information Technology — **2017-2019**

RESEARCH EXPERIENCE

Burmese News Summarization with Named Entity Recognition (NER)
Japan Advanced Institute of Technology (JAIST) - JAPAN

- Conducted research on **Burmese text summarization** by fine-tuning the **mT5 model** and evaluating the impact of **Named Entity Recognition (NER)** on summarization quality.
- Built an entity-tagging pipeline using **Conditional Random Fields (CRFs)** and **Stanza** for Burmese news articles and summaries.
- Implemented a **two-stage mT5 fine-tuning pipeline**, training first on entity-tagged data and subsequently refining the model with masked entities.
- Achieved a **4.06% improvement** over the latest fine-tuned Burmese mT5 summarization model.

CAPSTONE PROJECT - KMITL

Optical Network Resource Allocation Using Deep Reinforcement Learning

- Developed a full **Elastic Optical Network simulation** in Python using the COST239 topology with 11 nodes, 34 links, and 400 spectrum slots per link.
- Designed **Slicer-Aware PPO (SA-PPO)**, a two-stage deep reinforcement learning approach with a dynamic fragmentation-aware reward mechanism.
- Implemented and benchmarked **five spectrum allocation methods** using PyTorch and PPO: First-Fit, Conv-ML, FF-PPO, DRL-only, and SA-PPO.
- Reduced request blocking probability by **up to 80% at low load** and **approximately 40% at high load** compared with the First-Fit baseline.
- Developed visualization tools to inspect agent decisions and spectrum fragmentation.

HACKATHON PROJECT

AI Safety Assistant App - International AI Hackathon 2024 (Third Place)
Github : International AI Hackathon 2024

- Developed an **AI-powered workplace safety assistant** that analyzes **multimodal inputs** including images, voice, and text to support accident prevention and safety assessment.
- Implemented **OpenCV** and **YOLO** for image analysis and object detection to identify relevant visual safety information.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline using **Amazon Titan Embeddings** to retrieve relevant safety documents and provide context-aware information.
- Integrated **Claude 3 Sonnet** to generate safety recommendations, conversational responses, and accident analysis based on retrieved knowledge and user input.

ADDITIONAL PROJECTS

Automatic Pet Feeder - STM32F767ZI

Github : Embedded Systems Project

- Developed an automated pet feeder integrating **HC-SR04 ultrasonic** and **PIR sensors** for distance and motion detection.
- Implemented **timer-based input capture** and **stepper motor control** for automated food dispensing.
- Integrated **LCD and UART communication** for real-time system status, feeding notifications, and monitoring.

Cafe Shop Website - React

Github : React Web Application

- Developed **a React-based café ordering platform** with user accounts, menu browsing, item ordering, and order approval functionality.
- Built an **admin dashboard** for managing menus, categories, toppings, and incoming orders.
- Integrated APIs and **Axios** for client-server communication.

TECHNICAL SKILLS

- **Programming**: Python, C, JavaScript, PHP
- **AI / Machine Learning**: PyTorch, Transformers, mT5, PPO, LLMs, RAG, YOLO, OpenCV, CRF, Stanza, LangChain
- **Web Development**: React, Laravel, HTML, CSS, Tailwind CSS, Bootstrap, REST APIs
- **Database & Tools**: MySQL, Git, GitHub, Gradio, Streamlit
- **Embedded Systems**: STM32, Ultrasonic Sensors, PIR Sensors, Stepper Motors, UART, Timer/Input Capture

ACHIEVEMENTS

- **Bronze Award (Third Place) —** International AI Hackathon 2024, organized by KMITL, The University of Queensland, and NIA
- **Full Scholarship Recipient —** Bachelor's Degree in Computer Engineering, KMITL

LANGUAGES

English: Fluent | Myanmar: Native